Jiya Kaul

Technical Product Manager Assessment Submission | FactWise

Role: Technical Product Manager
Submission File: Jiya Kaul TPM.pdf
Includes Live Vercel Link & Codebase

Working Functional Prototype (Q2 Deliverable):
• **Live Deployed Link:** https://fact-wise-assessment-two.vercel.app
• **GitHub Repository:** https://github.com/Jiyakaul/FactWise-assessment

Problem Understanding (My Approach as a Product Manager)

The quarantine facility is facing an operational crisis: 74 patients (100% full occupancy), an aggressive virus with a 15% mortality benchmark, and severe staff burnout. After analyzing the head doctor's feedback, the root issues aren't lack of effort—they are **uncoordinated handoffs**:

- **Unsynchronized Rounds:** Doctors visit before nurses take vitals, wasting physician bandwidth on routine temperature measuring.
- **Discharge Bottlenecks:** Patients who meet the 3-day fever-free rule stay an extra 1–2 days because communication to the front desk is manual. This blocks waiting patients from entering the 74 beds.
- **Zero Performance Feedback:** The facility cannot see whether its treatment quality matches the 85% success benchmark.

Q1) App Users, Features, and Actions (Version 1 / MVP)

For the first version, I designed the product around 4 distinct personas so each person has a focused workspace with clear operational boundaries:

User Type	Problem Solved for This Role	Features & Permitted Actions
1. Nurse	Stops paper journal mix-ups, eliminates duplicate readings, and ensures every bed is checked daily.	• **Daily Rounding Queue:** Tabbed view of *Needs Check Today* vs *Completed Today*. • **Record Temperature Modal:** Direct entry in °F with instant color feedback (Green: <99.5°, Orange: 99.5–100.3°, Red: ≥100.4°F). • **Fever Streak Tracker:** Automatic badge showing consecutive fever-free progress (e.g., 2/3 Days). • *Action:* Record daily vitals and observations. Cannot authorize discharges or alter bed assignments.
2. Attending Doctor	Eliminates wasted visits where vitals are missing; provides clear 7-day vitals trends; automates cure sign-off.	• **Sequential Queue:** Segregated into *Ready for Round* (vitals already logged today) and *Pending Vitals* (warning indicator). • **Vitals Trendline Sparkline:** 7-day temperature curve with a clear 100.4°F fever reference line. • **Doctor Notes Log:** Document clinical evaluations and treatment changes. • *Action:* 1-click discharge sign-off (automatically enabled only when 3 fever-free days are verified).

User Type	Problem Solved for This Role	Features & Permitted Actions
3. Facility Admin / Front Desk	Keeps all 74 beds tracked in real-time, removes checkout lag, and prevents over-admission.	• **74-Bed Interactive Matrix:** Visual tiles for Beds 1–74 showing status at a glance (Red: Occupied, Amber: Discharge Pending, Green: Available). • **Discharge Desk:** Instant queue of doctor-approved discharges; 1-click "Finalize Discharge & Release Bed". • *Action:* Admit new patients into verified green beds; finalize departures; record patient mortality. Cannot edit clinical vitals.
4. Head Doctor / Medical Director	Gives leadership real-time visibility into whether the facility is meeting clinical benchmarks.	• **Quality Scorecards:** Live Mortality Rate vs the 15% benchmark and Cure Rate vs the 85% target. • **Critical Alert Banner:** Automatically flashes red if mortality rises above 15.0%. • **Shift Progress Telemetry:** Real-time progress bars tracking Nurse checks % and Doctor rounds % completed today. • *Action:* Monitor center-wide telemetry and average length of stay (ALOS).

Q2) Functional Prototype (Built & Deployed)

I built and deployed a production-grade web application to demonstrate the complete clinical workflow:

- **Live Deployed URL:** https://fact-wise-assessment-two.vercel.app
- **GitHub Repository:** https://github.com/Jiyakaul/FactWise-assessment
- **Tech Stack:** Next.js 14 App Router, React 18, TypeScript, Tailwind CSS, Recharts, and Lucide React.

> **How to Test the Key Scenarios in the Live App:**
>
> 1. **Nurse Streak Progression (Bed 5, 12, or 28):** These beds are pre-loaded with exactly 2 fever-free days. Go to Nurse View → Click "Record Temp" → Enter normal temp (e.g. 98.6°F) → Notice it advances to 3/3 days and triggers "Eligible for Discharge".
> 2. **Doctor Review & Discharge (Bed 40):** Bed 40 has completed 3 fever-free days. Switch to Doctor View → Open Bed 40 → The "Sign Discharge Order" button lights up green → Click to sign.
> 3. **Admin Bed Vacating (Bed 55):** Switch to Admin View → Discharge Desk → Click "Finalize Discharge & Release Bed" → Bed 55 turns green (Available) in the 74-bed matrix immediately.

Q3) Logical Constraints & Edge Cases for Developers

Beyond screens, developers need strict business logic so the system prevents human errors in an understaffed clinic:

1. Sequential Rounding Constraint (Nurse-First Rule)

- At `00:00:00` every night, daily vitals flags reset.
- If a doctor opens a patient whose nurse check is still pending, display an amber warning: "*Warning: Temperature has not been recorded today by a nurse.*"
- If the doctor decides to take the temperature anyway, allow manual intake right inside the modal, tag it as `recordedBy: "doctor"`, and clear the nurse's pending item so duplicate vitals aren't taken later.

2. The 3-Day Fever-Free Cure Rule (Strict Reset Engine)

- **Fever Definition:** Body temperature $T \ge 100.4^\circ\text{F}$ (38.0°C).
- **Strict Reset:** If a patient has 2 fever-free days and logs 100.5°F on Day 3, the counter **strictly resets to Day 0**. No exceptions.
- **Gated Sign-off:** The "Sign Discharge Order" button is hard-disabled until `consecutiveFeverFreeDays ≥ 3`.
- **Two-Stage Release:** Doctor approval sets status to `doctor_approved_discharge`. The bed is NOT vacated until Admin completes the checkout, ensuring front desk is never out of sync.

3. Capacity & Metric Integrity

- **Hard 74-Bed Limit:** New intake is disabled if active patients $= 74$. Trying to force-submit returns an error: "*Facility at maximum capacity (74/74 beds).*"
- **Closed Cases Formula:** To prevent distorting mortality with patients currently undergoing treatment, compute rates on closed outcomes: $$\text{Mortality Rate} = \frac{\text{Deceased}}

{\text{Discharged Cured} + \text{Deceased}} \times 100, \quad \text{Cure Rate} = \frac{\text{Discharged Cured}}{\text{Discharged Cured} + \text{Deceased}} \times 100$$

- **Benchmark Warning:** If Mortality Rate > 15.0%, trigger an alert banner across leadership dashboards.

Q4) Sprint Planning Mathematical Optimization (Knapsack Solution)

Team Parameters: Velocity = 25 Story Points (SP). Already committed = 3 stories × 5 SP = 15 SP. **Available Capacity = 10 SP**.
Value Rules: Low = 1 unit ($V_L = 1$), Medium = 3 units ($V_M = 3$), High = 6 units ($V_H = 6$).

Feature	Priority	Value (V)	Effort (W)	Value Density (V/W)	Decision
J	Low	1	0.1	**10.00**	**PICK (Top density)**
A	High	6	1.0	**6.00**	**PICK (High value, tiny effort)**
I	Low	1	0.3	**3.33**	**PICK**
F	Medium	3	1.0	**3.00**	**PICK**
G	Low	1	0.4	**2.50**	**PICK**
B	High	6	3.0	**2.00**	**PICK**
H	Low	1	0.5	**2.00**	**PICK**
D	Medium	3	2.0	**1.50**	**PICK**
C	High	6	5.0	1.20	SKIP (Consumes 50% of sprint capacity for 6 val)
E	Medium	3	3.0	1.00	SKIP (Lowest density)

Evaluating the Two Competing Options:

1. **The Intuitive "High-First" Approach {A, B, C, F}:**
   - Effort: $1 + 3 + 5 + 1 = \mathbf{10.0\text{ SP}}$ (100% capacity used, zero buffer)
   - Total Value Delivered: $6 + 6 + 6 + 3 = \mathbf{21\text{ Units}}$
2. **The Optimal Knapsack Selection {A, B, D, F, G, H, I, J}:**
   - Effort: $1.0 + 3.0 + 2.0 + 1.0 + 0.4 + 0.5 + 0.3 + 0.1 = \mathbf{8.3\text{ SP}}$
   - Total Value Delivered: $6 + 6 + 3 + 3 + 1 + 1 + 1 + 1 = \mathbf{22\text{ Units}}$
   - **Safety Buffer Remaining:** $10.0 - 8.3 = \mathbf{1.7\text{ Story Points}}$

> **My Decision as TPM:** Add **Features A, B, D, F, G, H, I, and J** to the sprint.
> Why? Feature C is a "trap" (5 points for only 6 value). By trading C for D and all low-effort items, we deliver **22 Value units instead of 21**, while using only **8.3 points instead of 10.0**. This leaves a 1.7-point (17%) buffer, which protects the team against unforeseen bugs or hospital emergencies.

Future Scope & Out-of-the-Box Product Ideas (v2 Roadmap)

To take this solution from a simple MVP to an enterprise-grade medical platform, here are key enhancements I would prioritize:

1. **Bedside Barcode / QR Wristband Scanning:** In an understaffed ward, nurses can easily confuse patients in neighboring beds. Equipping tablets with barcode scanners to scan patient

wristbands ensures 100% data integrity before logging vitals.

2. **Offline-First Progressive Web App (PWA):** Quarantine facilities often have thick concrete or shielded walls that drop Wi-Fi connectivity. Using service workers and local IndexedDB caching ensures nurses can log vitals completely offline; records sync automatically when back in Wi-Fi range.
3. **HIPAA / DISHA Compliant Role Permissions:** Ensure strict field-level masking. Nurses need bed numbers and vitals; admin staff need intake and billing; only licensed physicians should access full medical histories.
4. **Bluetooth BLE Smart Thermometer Integration:** Pairing the web app with contactless digital thermometers via Web Bluetooth API can auto-populate temperatures with zero typing, saving ~30 seconds per bed (37 minutes saved per nurse shift across 74 beds).